

Practical Assignment 5

Sr. No	Practical Question	Type
1	Given the list numbers = [10, 20, 30, 40, 50, 60, 70], use basic list slicing list[start:stop] to extract and print the sub-list [30, 40, 50].	Easy
2	Create a list alphabet = ["a", "b", "c", "d", "e", "f", "g", "h"]. Use slicing with a step value to print every second element starting from index 0.	Easy
3	Given the list data = [100, 200, 300, 400, 500], use slicing with a negative step [::-1] to create and print a reversed copy of the list.	Easy
4	Given scores = [88, 42, 95, 63, 71], write a program that uses the built-in functions min(), max(), and sum() to print the lowest score, highest score, and the total sum of all scores.	Easy
5	Create two lists: team_a = ["Alice", "Bob"] and team_b = ["Charlie", "David"]. Use the .extend() method to merge team_b into team_a. Print team_a.	Easy
6	Create a list votes = ["yes", "no", "yes", "yes", "no", "maybe"]. Use the .count() method to find and print how many times "yes" appears in the list.	Easy

7	Given cities = ["Mumbai", "Delhi", "Bangalore", "Chennai"], use the .index() method to find and print the position (index number) of "Bangalore".	Easy
8	Create a list prices = [450, 120, 890, 300, 50]. Use the .sort() method to sort the list in ascending order (lowest to highest) and print it.	Easy
9	Given marks = [67, 89, 45, 92, 78], use .sort(reverse=True) to sort the list in descending order (highest to lowest) and print it.	Mode rate
10	Create a list original = [1, 2, 3]. Use the .copy() method to create a duplicate named shallow_copy. Append 4 to shallow_copy and verify that original remains unchanged.	Mode rate
11	Create a list temp_logs = [24, 25, 26, 27]. Use the .clear() method to remove all elements from the list. Print the list and its length (len()) to verify it is empty.	Mode rate
12	Given a list matrix_row = [5, 10, 15, 20, 25, 30, 35, 40], use negative index slicing [-4:] to extract and print only the last 4 elements of the list.	Mode rate
13	Write a program that takes a user-entered word and converts it into a list of characters. Check if the list is equal to its slice reverse (lst == lst[::-1]) to determine if the word is a Palindrome (e.g., "radar"). Print True or False.	Mode rate

14	Given values = [12, 0, 45, 0, 78, 0, 90], write a program using a while loop and the .remove() method to delete all occurrences of 0 from the list. Print the cleaned list.	Mode rate
15	Create a list items = ["pen", "notebook", "eraser"]. Ask the user for an item name. Use an if-else statement paired with the in operator to check if it exists in the list. If it exists, print its index using .index(); if not, print "Item not found".	Mode rate

16. The Leaderboard Top-3 Filter

You are designing the post-match summary for an esports tournament. The game engine generates an unsorted list of player scores: all_scores = [340, 890, 120, 950, 670, 890, 410].

- Use the .sort() or .sort(reverse=True) method to organize the scores from highest to lowest.
- Use list slicing to extract only the **top 3 highest scores**.
- Calculate the average score of these top 3 players using sum() and len().
- Print the top 3 scores list and their calculated average score.

17. The E-Commerce Price Range Filter

An online shopping platform allows users to apply a price filter to search results. The system stores product prices in a list: product_prices = [15, 80, 120, 45, 300, 210, 95, 50].

- Ask the user to input two numbers: a min_price and a max_price.
- Write a program that iterates through product_prices and creates a new list named filtered_prices containing only the items that fall within [min_price, max_price] (inclusive).
- If no products match the criteria, use an if-else block to print "No items match your filter". Otherwise, print the sorted filtered_prices list using .sort().

18. The Playlist Re-orderer & Merger

A user is merging two music playlists on a streaming application.

- `rock_playlist = ["Bohemian Rhapsody", "Hotel California"]`
- `pop_playlist = ["Shape of You", "Blinding Lights", "Bad Guy"]`
- Merge `pop_playlist` into `rock_playlist` using `.extend()`.
- The user wants to insert a specific favorite track, "Stairway to Heaven", right into the middle of the newly combined playlist.
- Calculate the middle index using integer floor division (`len(list) // 2`).
- Use `.insert()` to place the track at that exact middle index and print the updated playlist.

19. The Duplicate Detector & Frequency Counter

A biometric attendance system registers card swipes by ID numbers in the order they occur: `swipes = [101, 102, 101, 103, 104, 101, 102, 105]`.

- Create an empty list named `unique_swipes`.
- Loop through each ID in `swipes`. If the ID is not already in `unique_swipes`, add it using `.append()`.
- After building the unique list, iterate through `unique_swipes` and use `.count()` on the original `swipes` list to display how many times each employee swiped their card (*e.g., "ID 101 swiped 3 times"*).

20. The Security Log Trim (Tail Analysis)

A cloud security server generates system log entries continuously: `logs = ["LOG1", "LOG2", "LOG3", "LOG4", "LOG5", "LOG6", "LOG7", "LOG8", "LOG9", "LOG10"]`. Due to memory limits, the system administrator only wants to keep the **most recent 5 log entries**.

- Use negative slicing to extract the last 5 logs into a new variable `recent_logs`.
- The administrator needs these 5 recent logs displayed in **reverse chronological order** (most recent first). Reverse the `recent_logs` list using the `.reverse()` method.
- Loop through the reversed list and print each log entry prefixed with "CRITICAL AUDIT: " using a for loop.